

ELEC4631 Tutorial 10

1. Consider the LTI SISO system

$$S = \left(\begin{bmatrix} -5 & 3 \\ -2 & 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, [4 \quad 3], 2 \right).$$

Find a state-feedback control signal of the form $u(t) = -Kx(t) + Nr(t)$ that will put the poles of the system at $(-3 + j, -3 - j)$, and force the output $y(t)$ to follow any step input $r(t)$ at steady state.

2. Solving the equation $\gamma(s) = \det(sI - A + BK + LC - \beta K)$ in a brute-force manner for β as in the example in Lecture 13 for given $\gamma(s)$, A , B , C , and L , and K , can be a tedious process. But it's possible to be a bit clever about this and proceed in a more efficient way.

(a) Suppose that $(K, A - LC)$ is an observable pair, show that by exploiting Ackermann's Formula one can compute a matrix X such that $\beta = B - X$.

(b) Suppose that $A = \begin{bmatrix} -5 & 3 \\ -2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $C = [4 \quad 3]$, $K = [1 \quad -2]$, and $L = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Use the technique in Part (a) to solve the equation

$$\det(sI - A + BK + LC - \beta K) = s^2 + 6s + 25.$$

for β .

3. Consider a system with transfer function

$$H(s) = \frac{3(s+7)(s+2)}{(s+1)(s+3)(s+14)}$$

(a) Let $E(s)$ be a rational transfer function with at most one pole at the origin and with all other poles in the LHP. The Final Value Theorem of the Laplace transform then states that $\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s)$.

Now you will recall a basic result from a first control systems course (e.g., ELEC3114). Let $H(s)$ be a transfer function with all its poles on the LHP. Show that if $H(s)$ has dc gain 1 ($H(0) = 1$) then the output of the system will follow a step input with no error.

- (b) Verify that the system has DC gain 1.
- (c) Using Truxal's Formula, determine the steady-state error of the system in following a ramp input.
- (d) Find the Type 1 system $G(s)$ so that $H(s)$ can be expressed as $H(s) = \frac{G(s)}{1+G(s)}$.
- (e) Compute K_v from $G(s)$ and compare it with the value obtained using Truxal's Formula in part (b).

4. Consider the state-space system

$$\begin{aligned} \dot{x} &= \begin{bmatrix} 2 & 0 \\ -2 & 1 \end{bmatrix} x + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u, \\ y &= \begin{bmatrix} 0 & 1 \end{bmatrix} x. \end{aligned}$$

- (a) Show that this system will not be able to track a step input signal for any state-feedback gain K .
- (b) Compute the transfer function for this system. What is distinctive about it? Use this to explain why the system is unable to track a step signal (**hint:** Examine the closed-loop zeros under any stabilizing feedback K).

5. Given the second order state-space system $S = (A, B, C, 0)$ with

$$A = \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}; B = \begin{bmatrix} 6 \\ 4 \end{bmatrix}; C = \begin{bmatrix} 0 & 1 \end{bmatrix}, D = 0.$$

- (a) What is the system transfer function?
- (b) What is the system type?
- (c) Design an output feedback controller with a zero assignment estimator such that the compensated system is able to track a step input at steady state, has dominant closed-loop poles at $p = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$ with damping ratio $\zeta = 0.8$ and natural frequency $\omega_n = 5$, and has $K_v = 20$. Show clearly each step of your design (**Hint:** You may want to make life easier for yourself by using the result from Problem 2).
- (d) Write down the resulting closed-loop transfer function from input to output.
- (e) Plot the step and ramp response of the compensated system from part (c) in Matlab.